

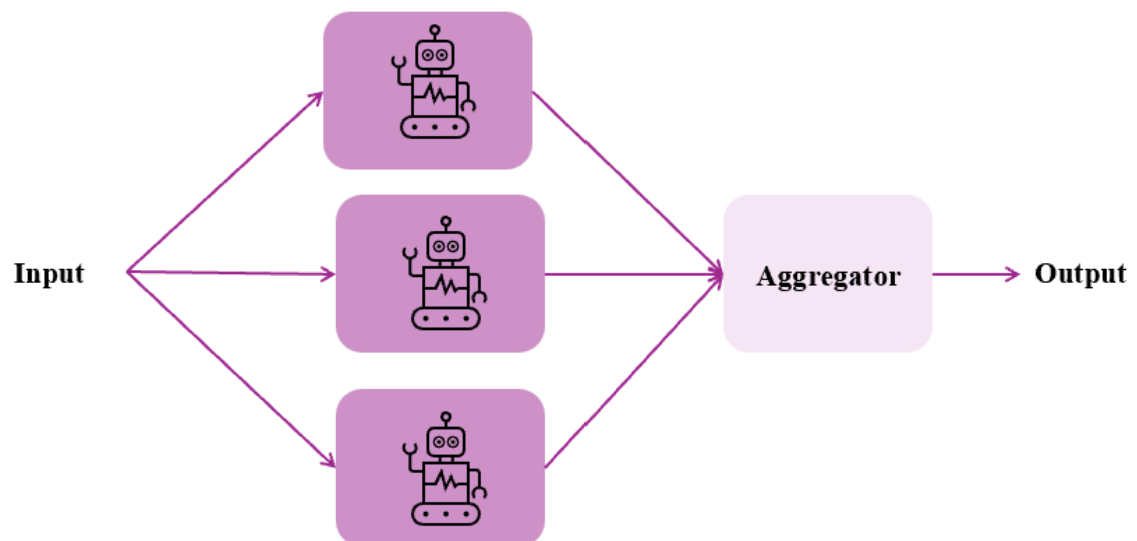
Agentic AI Framework Types

Parallelization

What is Parallelization in LangGraph ?

In LangGraph, nodes typically execute in a sequence defined by the edges, but when don't depend on each other's outputs, you can run them in parallel. This is achieved by :

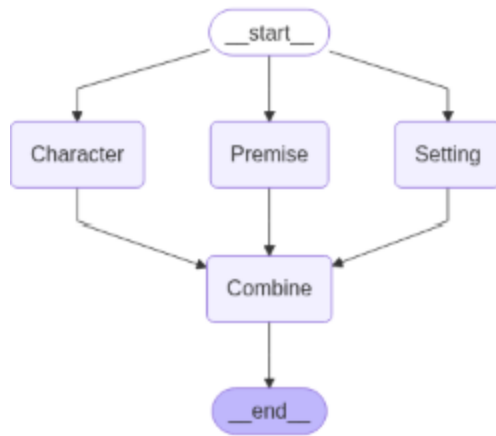
- Defining multiple nodes that can operate independently.
- Connecting them to a common starting point (eg. START Or another node).
- Merging their outputs into a downstream node if needed.



Implementation

Aim :

Write a story by generating character, premise, setting for the story parallelly and then combine them to write a introduction to the story.



Step 1 : Importing Libraries

```

In [1]: import os
        from dotenv import load_dotenv
        load_dotenv()

        from typing_extensions import TypedDict
        from langgraph.graph import StateGraph, START, END
        from langchain_openai import ChatOpenAI
        from IPython.display import Image, display
  
```

Step 2 : Create the LLM

```

In [2]: llm = ChatOpenAI(model="gpt-4o")
  
```

Step 3 : Create State Schema

```

In [9]: class State(TypedDict):
        topic:str
        characters:str
        settings:str
        premises:str
        story_intro:str
  
```

Step 4 : Create the Function for the node

```

In [10]: # Node 1 : Generate Character descriptions
        def generate_character(state:State):
            msg = llm.invoke(f"Create two character names and brief traits for a story about {state['topic']}")
            return {"characters":msg.content}

        # Node 2 : Generate a story setting
        def generate_setting(state:State):
            msg = llm.invoke(f"Describe a vivid setting for a story about {state['topic']}")
            return {"settings":msg.content}
  
```

```

# Node 3 : Generate a story premise
def generate_premise(state:State):
    msg = llm.invoke(f"Write a one-sentence plot premise for a story about {state['characters']}")
    return {"premises":msg.content}

# Node 4 combine Characters, settings, and premises into an intro
def combine_elements(state:State):
    msg = llm.invoke(
        f"Write a short story introduction using these elements:\n"
        f"Characters: {state['characters']}\n"
        f"Setting: {state['settings']}\n"
        f"Premise: {state['premises']}\n"
    )
    return {"story_intro":msg.content}

```

Step 5 : Build the Complete Graph

```

In [11]: builder = StateGraph(State)

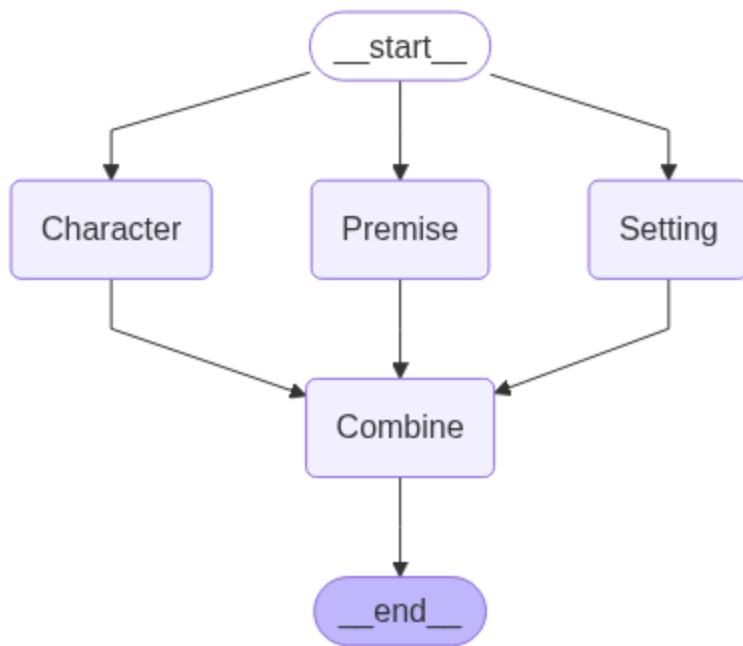
# Add the nodes
builder.add_node("Character", generate_character)
builder.add_node("Setting", generate_setting)
builder.add_node("Premise", generate_premise)
builder.add_node("Combine", combine_elements)

# Add the edge
builder.add_edge(START, "Character")
builder.add_edge(START, "Setting")
builder.add_edge(START, "Premise")
builder.add_edge("Character", "Combine")
builder.add_edge("Setting", "Combine")
builder.add_edge("Premise", "Combine")
builder.add_edge("Combine", END)

# Compile the entire Graph
graph = builder.compile()

# View the Entire graph
display(Image(graph.get_graph().draw_mermaid_png()))

```



Step 6 : Graph Invocation

```
In [12]: state={"topic":"Time Travel"}
result = graph.invoke(state)
result
```

```
Out[12]: {'topic': 'Time Travel',
          'characters': "Certainly! Here are two character names with brief traits for your
time travel story:\n\n1. **Evelyn Harper** \n - **Occupation**: Quantum Physi-
cist \n - **Traits**: Evelyn is brilliant, analytical, and driven by an insatiabl-
e curiosity about the mysteries of time. She has a quirky sense of humor and often
speaks in technical jargon, but she can simplify complex ideas when required. Her
empathy makes her deeply concerned about the ethical implications of time travel,
and she constantly grapples with the potential consequences of altering timeline-
s.\n\n2. **Finn Mercer** \n - **Occupation**: History Enthusiast and Adventurer
\n - **Traits**: Finn is charismatic, adventurous, and a bit of a risk-taker. Wi-
th a background in archaeology, he's passionate about experiencing historical even-
ts firsthand and learning from them in a way books can't teach. He tends to rely o-
n his instincts, occasionally clashing with Evelyn's methodical approach. Despite
his laid-back demeanor, Finn is fiercely protective of history's authenticity and
passionate about preserving the natural course of events.",
          'settings': 'In the heart of a sprawling metropolis, hidden beneath the bustling
streets and towering skyscrapers, lies the enigmatic Sanctum of Eternity—a clandes-
tine hub for time travelers known only to a select few. The entrance, concealed wi-
thin the aged façade of an unassuming antique clock shop, is obscured by the rhyth-
mic ticking of countless timepieces from eras long past. \n\nAs one steps inside,
the atmosphere shifts, and the air is tinged with the faint scent of polished bras-
s and aged paper. Dimly lit by flickering Edison bulbs, the shop reveals itself as
a maze of narrow aisles lined with clock faces, hourglasses, and intricate gears,
each seemingly alive, silently ticking away in a cosmic synchronization.\n\nPushin-
g through a heavy oak door at the back, one enters the Sanctum's nerve center—a ca-
vernous underground chamber sprawling with anachronistic charm. Here, polished mar-
ble floors reflect a kaleidoscope of colors from the grand stained-glass dome abov-
e, depicting constellations of the universe intertwined with symbols of time.\n\nT-
he chamber is filled with the soft hum of technology, where futuristic holographic
screens flicker beside Victorian control panels of brass and mahogany. A massive c-
entral console, adorned with shimmering quartz and arcane runes, pulses with a gen-
tle azure light—the core of the time travel apparatus.\n\nEncircling the console,
a ring of tall stone arches stands, each inscribed with cryptic alphabets and glow-
ing faintly, resonating with the echoes of countless journeys through time. These
are the Vortex Gates, each portal a gateway to a different epoch, spinning open wi-
th a whispered command, inviting adventurers to step through the veils of time.\n
\nAmidst this fusion of antiquity and advanced technology, scholars huddle togethe-
r, poring over ancient scrolls and futuristic tablets alike. They debate theories
and predict paradoxes in hushed tones as travelers don meticulously crafted attire
for their chosen destinations, adjusting intricate devices strapped to their wrist-
s.\n\nThe chamber vibrates with an aura of mystery and possibility—a place where h-
istory and the unknown converge, where the air crackles with the promise of advent-
ure across the ages and the whisper of past, present, and future converging into o-
ne. Here, the veil of time is but a fragile curtain, ready to be drawn aside for t-
hose bold enough to take the leap.',
          'premises': 'In a world where time travel is forbidden, a rebellious historian di-
scovers an ancient bracelet that propels her into the past, unraveling her own tan-
gled ancestry as she races against time to prevent a catastrophic event.',
          'story_intro': 'Evelyn Harper adjusted her glasses, the lenses gleaming under the
tungsten glow of the Sanctum of Eternity's hidden heart. She weaved through the m-
aze of clocks and hourglasses, their tick-tocking a timeless symphony, as she appr-
oached the cavernous underground chamber. Here, time breathed with a life of its o-
wn, a merging of epochs where history's heartbeat thrummed beneath polished marble
floors.\n\nAt the center of the chamber, Finn Mercer stood, his silhouette cast ag-
ainst the kaleidoscopic dance of stained glass above. He traced the symbols on the
stone arches with a cautious reverence, his expression one of awe and mischief—a m
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an caught between wonder and an unquenchable thirst for adventure.\n\n"Finn, you\n're daydreaming again," Evelyn teased, her voice mingling with the hum of arcane machinery that reverberated through the walls.\n\nFinn flashed a roguish grin, his eyes twinkling like the constellations captured above. "Just imagining the stories these stones could tell, Evelyn. A thousand lifetimes waiting to be explored."\n\n"Yet only one we should focus on," Evelyn countered, her gaze drifting to the massive central console, its shimmering quartz pulsating with an ethereal glow. "Time is not to be trifled with, especially when we're on the brink of a discovery that defies our own rules."\n\nFinn nodded, his adventurous spirit tempered by the weight of responsibility. "I know, but someone has to make sure history's not repeating—or worse, unraveling."\n\nThe air in the chamber seemed to tighten as Evelyn's thoughts drifted to the ancient bracelet she had unwittingly uncovered. Its presence was an enigma, a conundrum of temporal dimensions she couldn't fully grasp, yet Finn seemed undeterred by the impending complications it might bring.\n\n"Do you think..." Finn began, his voice a tentative whisper shared only between the two of them, "that this bracelet could be the key to untangling your lineage, and maybe even ours?"\n\nEvelyn hesitated, the gravity of the situation settling like a shadow over her features. "It might be," she admitted. "But it could also lead to unimaginable chaos if we're not careful. And in a world where time travel is forbidden, even being here is a risk."\n\nFinn laughed, a sound both rebellious and reassuring. "What's life without a little risk? Besides," he added, gesturing to the swirling Vortex Gates, "it's not every day you get a chance to make history a part of yourself."\n\nShe couldn't help but smile at his indomitable spirit, a contrast to her cautious approach, yet strangely harmonious in their shared quest. Together, they stood at the precipice of time itself, ready to unearth the secrets hidden within their own pasts and protect the fragile dance of history from the precipice of catastrophe.\n\nSide by side, Evelyn and Finn faced the whispers of eternity, the air thick with possibility. As Evelyn placed the bracelet on her wrist, she felt the resonance of time vibrating through her very being—a promise of adventure and the weight of consequence intertwined—in a leap through the veils of time.'}

Key Benefits:

- Speed : Reduces total execution time by running tasks concurrently
- Scalability : Handles larger workflows efficiently
- Modularity : Keeps the graph structure clean and reusable.

Key Takeaways:

- When to Parallelize : Use it for independent tasks(eg. generating multiple outputs, checking separate inputs).
- Merging : Downstream nodes can aggregate parallel results.
- LangGraph Support : The framework naturally supports this by waiting for all required inputs before proceeding.

In []: